N2 Normal distribution A level only

Name: ________________

Class: ________________

Date: ________________

Time: **434 minutes**

Marks: **357 marks**

Comments:

Q1.

A survey of 120 adults found that the volume, X litres per person, of carbonated drinks they consumed in a week had the following results:

$$\sum x = 165.6 \qquad \sum x^2 = 261.8$$

(a) (i) Calculate the mean of X.

(1)

(ii) Calculate the standard deviation of X.

(2)

(b) Assuming that X can be modelled by a normal distribution find

(i) P(0.5 < X < 1.5)

(2)

(ii) P(X = 1)

(1)

(c) Determine with a reason, whether a normal distribution is suitable to model this data.

(2)

(d) It is known that the volume, Y litres per person, of energy drinks consumed in a week may be modelled by a normal distribution with standard deviation 0.21

Given that P(Y > 0.75) = 0.10, find the value of μ, correct to three significant figures.

(4)

(Total 12 marks)

Q2.

In the South West region of England, 100 households were randomly selected and, for each household, the weekly expenditure, £X , per person on food and drink was recorded.

The maximum amount recorded was £40.48 and the minimum amount recorded was £22.00
The results are summarised below, where x̄ denotes the sample mean.

$$\sum x = 3046.14 \qquad \sum (x - \bar{x})^2 = 1746.29$$

(a) (i) Find the mean of X
Find the standard deviation of X

(2)

(ii) Using your results from part **(a)(i)** and other information given, explain why the normal distribution can be used to model X.

(2)

(iii) Find the probability that a household in the South West spends less than £25.00 on food and drink per person per week.

(1)

(b) For households in the North West of England, the weekly expenditure, £Y, per person on food and drink can be modelled by a normal distribution with mean £29.55

It is known that P(Y < 30) = 0.55

Find the standard deviation of Y, giving your answer to one decimal place.

(3)

(Total 8 marks)

Q3.

A survey during 2013 investigated mean expenditure on bread and on alcohol.

The 2013 survey obtained information from 12 144 adults.
The survey revealed that the mean expenditure per adult per week on bread was 127p.

(a) For 2012, it is known that the expenditure per adult per week on bread had mean 123p, and a standard deviation of 70p.

(i) Carry out a hypothesis test, at the 5% significance level, to investigate whether the mean expenditure per adult per week on bread changed from 2012 to 2013.

Assume that the survey data is a random sample taken from a normal distribution.

(5)

(ii) Calculate the greatest and least values for the sample mean expenditure on bread per adult per week for 2013 that would have resulted in acceptance of the null hypothesis for the test you carried out in part **(a)(i)**.

Give your answers to two decimal places.

(2)

(b) The 2013 survey revealed that the mean expenditure per adult per week on alcohol was 324p.

The mean expenditure per adult per week on alcohol for 2009 was 307p.

A test was carried out on the following hypotheses relating to mean expenditure per adult per week on alcohol in 2013.

$H_0 : \mu = 307$

$H_1 : \mu \neq 307$

This test resulted in the null hypothesis, H_0, being rejected.

State, with a reason, whether the test result supports the following statements:

(i) the mean UK expenditure on alcohol per adult per week increased by 17p from 2009 to 2013;

(2)

(ii) the mean UK consumption of alcohol per adult per week changed from 2009 to 2013.

(2)

(Total 11 marks)

Q4.

The volume, V litres, of *Cleanall* washing-up liquid in a 5-litre container may be modelled by a normal distribution with a mean, μ, of 5.028 and a standard deviation of 0.015.

(a) Determine the probability that the volume of *Cleanall* in a randomly selected 5-litre container is:

(i) less than 5.04 litres;

(ii) more than 5 litres.

(5)

(b) Determine the value of v such that $\mathrm{P}(\mu - v < V < \mu + v) = 0.95$.

(4)

(Total 9 marks)

Q5.

The volume of *Everwhite* toothpaste in a pump-action dispenser may be modelled by a normal distribution with a mean of 106 ml and a standard deviation of 2.5 ml.

Determine the probability that the volume of *Everwhite* in a randomly selected dispenser is:

(a) less than 110 ml;

(3)

(b) more than 100 ml;

(2)

(c) between 104 ml and 108 ml;

(3)

(d) **not** exactly 106 ml.

(1)

(Total 9 marks)

Q6.

A machine, which cuts bread dough for loaves, can be adjusted to cut dough to any specified set weight. For any set weight, μ grams, the actual weights of cut dough are known to be approximately normally distributed with a mean of μ grams and a fixed standard deviation of σ grams.

It is also known that the machine cuts dough to within 10 grams of any set weight.

(a) Estimate, with justification, a value for σ.

(2)

(b) The machine is set to cut dough to a weight of 415 grams.

As a training exercise, Sunita, the quality control manager, asked Dev, a recently employed trainee, to record the weight of each of a random sample of 15 such pieces of dough selected from the machine's output. She then asked him to calculate the mean and the standard deviation of his 15 recorded weights.

Dev subsequently reported to Sunita that, for his sample, the mean was 391 grams and the standard deviation was 95.5 grams.

Advise Sunita on whether or not **each** of Dev's values is likely to be correct. Give numerical support for your answers.

(3)

(c) Maria, an experienced quality control officer, recorded the weight, y grams, of each of a random sample of 10 pieces of dough selected from the machine's output when it was set to cut dough to a weight of 820 grams. Her summarised results were as follows.

$$\sum y = 8210.0 \quad \text{and} \quad \sum (y - \bar{y})^2 = 110.00$$

Explain, with numerical justifications, why **both** of these values are likely to be correct.

(4)

(Total 9 marks)

Q7.

The volume, X millilitres, of energy drink in a bottle can be modelled by a normal random variable with mean 507.5 and standard deviation 4.0.

(a) Find:

(i) P(X < 515);

(ii) P(500 < X < 515);

(iii) P($X \neq$ 507.5).

(5)

(b) Determine the value of x such that P(X < x)= 0.96.

(3)

(c) The energy drink is sold in packs of 6 bottles. The bottles in each pack may be regarded as a random sample.

Calculate the probability that the volume of energy drink in at least 5 of the 6 bottles in a pack is between 500 ml and 515 ml.

(3)
(Total 11 marks)

Q8.

The weight, X grams, of the contents of a tin of baked beans can be modelled by a normal random variable with a mean of 421 and a standard deviation of 2.5.

(a) Find:

(i) $P(X = 421)$;

(ii) $P(X < 425)$;

(iii) $P(418 < X < 424)$.

(6)

(b) Determine the value of x such that $P(X < x) = 0.98$.

(3)

(c) The weight, Y grams, of the contents of a tin of ravioli can be modelled by a normal random variable with a mean of μ and a standard deviation of 3.0.

Find the value of μ such that $P(Y < 410) = 0.01$.

(4)
(Total 13 marks)

Q9.

To supplement her pension, Cordelia bakes cakes which are then sold in Amir's nearby shop.

(a) The weight, X grams, of her square fruit cake may be modelled by a normal random variable with a mean of 2200 and a standard deviation of 160.

Determine:

(i) $P(X < 2500)$;

(ii) $P(X > 2000)$;

(iii) $P(2000 < X < 2500)$.

(7)

(b) The weight, Y grams, of Cordelia's round fruit cake may be modelled by a normal random variable with a mean of 1125 and a standard deviation of σ.

If Amir requests that at most 10 per cent of these cakes should weigh less than 1000 grams, find the maximum value for σ.

(4)
(Total 11 marks)

Q10.

A sawmill's machine cuts stakes for use in constructing agricultural fences. The lengths of the cut stakes may be modelled approximately by a normal distribution.

The machine is set to cut stakes of length 1.5 m and is known to cut stakes to within 10 cm of any set length.

Elmer, the sawmill's foreman, asked Ashley, a trainee office junior, to calculate the mean and the standard deviation for the lengths of a random sample of 10 such stakes selected from the machine's output.

Ashley reported back to Elmer that the mean length of his 10 selected stakes was 151.5 cm and that the standard deviation of their lengths was 26.6 cm.

Advise Elmer on whether **each** of Ashley's values could be correct. Give numerical support for your answers.

(4)
(Total 4 marks)

Q11.

During June 2011, the volume, X litres, of unleaded petrol purchased per visit at a supermarket's filling station by private-car customers could be modelled by a normal distribution with a mean of 32 and a standard deviation of 10.

(a) Determine:

(i) $\mathrm{P}(X < 40)$;

(ii) $\mathrm{P}(X > 25)$;

(iii) $\mathrm{P}(25 < X < 40)$.

(7)

(b) Given that during June 2011 unleaded petrol cost £1.34 per litre, calculate the probability that the unleaded petrol bill for a visit during June 2011 by a private-car customer exceeded £65.

(3)

(c) Give **two** reasons, in context, why the model $\mathrm{N}(32, 10^2)$ is unlikely to be valid for a visit by **any** customer purchasing fuel at this filling station during June 2011.

(2)
(Total 12 marks)

Q12.

A general store sells lawn fertiliser in 2.5 kg bags, 5 kg bags and 10 kg bags.

(a) The actual weight, W kilograms, of fertiliser in a 2.5 kg bag may be modelled by a normal random variable with mean 2.75 and standard deviation 0.15.

Determine the probability that the weight of fertiliser in a 2.5 kg bag is:

(i) less than 2.8 kg;

(ii) more than 2.5 kg.

(5)

(b) The actual weight, X kilograms, of fertiliser in a 5 kg bag may be modelled by a normal random variable with mean 5.25 and standard deviation 0.20.

(i) Show that $\mathrm{P}(5.1 < X < 5.3) = 0.372$, correct to three decimal places.

(2)

(ii) A random sample of **four** 5 kg bags is selected. Calculate the probability that none of the four bags contains between 5.1 kg and 5.3 kg of fertiliser.

(2)

(Total 9 marks)

Q13.

The weight, W grams, of a bag of sugar may be assumed to be normally distributed with a mean of 1018 and a standard deviation of 10.

Determine the probability that:

(a) W is less than 1025;

(2)

(b) W is greater than 1015 but less than 1030.

(3)

(Total 5 marks)

Q14.

Craig uses his car to travel regularly from his home to the area hospital for treatment. He leaves home at x minutes after 7.30 am and then takes y minutes to arrive at the hospital's reception desk.

His results for 11 mornings are shown in the table.

x	0	5	10	15	20	25	30	35	40	45	50
y	31	42	32	58	47	56	79	68	89	95	85

(a) Explain why the time taken by Craig between leaving home and arriving at the hospital's reception desk is the response variable.

(1)

(b) Calculate the equation of the least squares regression line of y on x, writing your answer in the form $y = a + bx$.

(5)

(c) On a particular day, Craig needs to arrive at the hospital's reception desk no later than 9.00 am. He leaves home at 7.45 am.

Estimate the number of minutes **before** 9.00 am that Craig will arrive at the hospital's reception desk. Give your answer to the nearest minute.

(5)

(d) (i) Use your equation to estimate y when x = 85.

(1)

(ii) Give **one** statistical reason and one reason based on the context of this question as to why your estimate in part **(d)(i)** is unlikely to be realistic.

(2)

(Total 14 marks)

Q15.

The volume of shampoo, V millilitres, delivered by a machine into bottles may be modelled by a normal random variable with mean μ and standard deviation σ.

(a) Given that μ = 412 and σ = 8, determine:

(i) P(V < 400);

(3)

(ii) P(V > 420);

(2)

(iii) P(V = 410).

(1)

(b) A new quality control specification requires that the values of μ and σ are changed so that

P(V < 400) = 0.05 and P(V > 420) = 0:01

(i) Show, with the aid of a suitable sketch, or otherwise, that

400 – μ = –1:6449μ and 420 – μ = 2:3263σ

(3)

(ii) Hence calculate values for μ and σ.

(3)

(Total 12 marks)

Q16.

A machine produces ice hockey pucks whose weights may be modelled by a normal distribution with a mean of 165 grams and a standard deviation of σ grams.

(a) Given that $\sigma = 2.5$, determine the probability that the weight of a randomly selected puck is:

(i) less than 167 grams;

(3)

(ii) more than 162 grams.

(2)

(b) An ice hockey club purchases a box of 12 pucks produced by the machine.

Assuming that the pucks in any box represent a random sample, calculate the probability that all 12 pucks weigh less than 167 grams.

(2)

(c) An ice hockey confederation requires that at most 1 per cent of pucks have weights outside the range 160 grams to 170 grams.

Assuming that the value of the mean remains unchanged at 165 grams, calculate, to two decimal places, the maximum value of s which meets this requirement.

(4)

(Total 11 marks)

Q17.

The diameter, D millimetres, of an American pool ball may be modelled by a normal random variable with mean 57.15 and standard deviation 0.04.

(a) Determine:

(i) P(D < 57:2);

(3)

(ii) P(57.1 < D < 57.2).

(2)

(b) A box contains 16 of these pool balls. Given that the balls may be regarded as a random sample, determine the probability that all 16 balls have diameters less than 57.2 mm;

(2)

(Total 7 marks)

Q18.

Draught excluder for doors and windows is sold in rolls of nominal length 10 metres.

The actual length, X metres, of draught excluder on a roll may be modelled by a normal distribution with mean 10.2 and standard deviation 0.15.

(a) Determine:

(i) P(X < 10.5);

(3)

(ii) P(10.0 < X < 10.5).

(3)

(b) A customer randomly selects six 10-metre rolls of the draught excluder.

Calculate the probability that all six rolls selected contain more than 10 metres of draught excluder.

(3)
(Total 9 marks)

Q19.

A machine fills small cans with soda water. The volume of soda water delivered by the machine may be modelled by a normal random variable with a mean of 153 ml and a standard deviation of 1.6 ml.

Each can is able to hold a maximum of 155 ml of soda water.

Printed on each can is 'Contents 150 ml'.

Determine the probability that the volume of soda water delivered by the machine:

(a) does not cause a can to overflow;

(b) is less than that printed on a can.

(Total 6 marks)

Q20.

Each day, Margot completes the crossword in her local morning newspaper. Her completion times, X minutes, can be modelled by a normal random variable with a mean of 65 and a standard deviation of 20.

(a) Determine:

(i) P(X < 90);

(ii) P(X > 60).

(5)

(b) Given that Margot's completion times are independent from day to day, determine the probability that, during a particular period of 6 days:

(i) she completes one of the six crosswords in exactly 60 minutes;

(1)

(ii) she completes each crossword in less than 60 minutes;

(3)
(Total 9 marks)

Q21.

The quarterly consumption, X m^3, of water by two-person households may be modelled by a normal distribution with mean μ and variance σ^2.

From an analysis of records, it is found that $P(X < 45) = 0.98$.

(a) Show that

$$45 - \mu = 2.0537\sigma$$

(3)

(b) Given also that $P(X > 30) = 0.95$, find, to one decimal place, values for μ and σ.

(4)

(Total 7 marks)

Q22.

UPVC facia board is supplied in lengths labelled as 5 metres. The actual length, X metres, of a board may be modelled by a normal distribution with a mean of 5.08 and a standard deviation of 0.05.

(a) Determine:

(i) $P(X < 5)$;

(3)

(ii) $P(5 < X < 5.10)$.

(2)

(b) Assuming that the value of the standard deviation remains unchanged, determine the mean length necessary to ensure that only 1 per cent of boards have lengths less than 5 metres.

(4)

(Total 9 marks)

Q23.

A machine fills boxes with wine. The volume, W litres, of wine delivered by the machine into a box may be modelled by a normal distribution with mean 3.12 and standard deviation σ.

(a) Given that $\sigma = 0.08$, determine $P(2.95 < W < 3.20)$.

(4)

(b) Assuming that the value of the mean remains unchanged, determine the value of σ necessary to ensure that at most 2.5% of boxes filled by the machine contain less than 3 litres of wine.

(4)

(Total 8 marks)

Q24.

The weight, X grams, of talcum powder in a tin may be modelled by a normal distribution with mean 253 and standard deviation σ.

(a) Given that σ = 5, determine:

(i) P(X < 250);

(3)

(ii) P(245 < X < 250);

(2)

(iii) P(X = 245).

(1)

(b) Assuming that the value of the mean remains unchanged, determine the value of σ necessary to ensure that 98% of tins contain more than 245 grams of talcum powder.

(4)
(Total 10 marks)

Q25.

In large-scale tree-felling operations, a machine cuts down trees, strips off the branches and then cuts the trunks into logs of length X metres for transporting to a sawmill.

It may be assumed that values of X are normally distributed with mean 3.3 and standard deviation 0.16.

Determine:

(a) P(X < 3.5);

(3)

(b) P(X > 3.0);

(3)

(c) P(3:0 < X < 3:5).

(2)
(Total 8 marks)

Q26.

In large-scale tree-felling operations, a machine cuts down trees, strips off the branches and then cuts the trunks into logs of length X metres for transporting to a sawmill.

It may be assumed that values of X are normally distributed with mean μ and standard deviation 0.16, where μ can be set to a specific value.

(a) Given that μ is set to 3.3, determine:

(i) P(X < 3.5);

(3)

(ii) P(X > 3.0);

(3)

(iii) P(3.0 < X < 3.5).

(2)

(b) The sawmill now requires a batch of logs such that there is a probability of 0.025 that any given log will have a length less than 3.1 metres.

Determine, to two decimal places, the new value of μ.

(4)
(Total 12 marks)

Q27.

The length, L centimetres, of *Slimline* bin liners may be modelled by a normal distribution with a mean of 69.5 and a standard deviation of 0.55.

(a) Determine:

(i) P(L < 70);

(3)

(ii) P(69 < L < 70);

(3)

(iii) P(L = 70).

(1)

(b) Determine the maximum length exceeded by 90% of bin liners.

(4)

(c) The bin liners are sold in packets of 20, and those in each packet may be considered to be a random sample.

Determine the probability that the mean length of the bin liners in a packet is greater than 69.25 cm.

(4)
(Total 15 marks)

Q28.

When a particular make of tennis ball is dropped from a vertical distance of 250 cm on to concrete, the height, X centimetres, to which it first bounces may be assumed to be normally distributed with a mean of 140 and a standard deviation of 2.5.

(a) Determine:

(i) P(X < 145);

(3)

(ii) P(138 < X < 142).

(4)

(b) Determine, to one decimal place, the maximum height exceeded by 85% of first bounces.

(4)

(Total 11 marks)

Q29.

When Monica walks to work from home, she uses either route A or route B.

(a) Her journey time, X minutes, by route A may be assumed to be normally distributed with a mean of 37 and a standard deviation of 8.

Determine:

(i) P(X < 45);

(3)

(ii) P(30 < X < 45).

(3)

(b) Her journey time, Y minutes, by route B may be assumed to be normally distributed with a mean of 40 and a standard deviation of σ.

Given that P(Y > 45) = 0.12, calculate the value of σ.

(4)

(c) If Monica leaves home at 8.15 am to walk to work hoping to arrive by 9.00 am, state, with a reason, which route she should take.

(2)

(Total 12 marks)

Q30.

When Monica walks to work from home, she uses either route A or route B.

(a) Her journey time, X minutes, by route A may be assumed to be normally distributed with a mean of 37 and a standard deviation of 8.

Determine:

(i) P(X < 45);

(3)

(ii) P(30 < X < 45).

(3)

(b) Her journey time, Y minutes, by route B may be assumed to be normally distributed with a mean of 40 and a standard deviation of σ.

Given that P(Y > 45) = 0.12, calculate the value of σ.

(4)

(c) If Monica leaves home at 8.15 am to walk to work hoping to arrive by 9.00 am, state, with a reason, which route she should take.

(2)

(d) When Monica travels to work from home by car, her journey time, W minutes, has a mean of 18 and a standard deviation of 12.

Estimate the probability that, for a random sample of 36 journeys to work from home by car, Monica's mean time is more than 20 minutes.

(4)

(e) Indicate where, if anywhere, in this question you needed to make use of the Central Limit Theorem.

(1)

(Total 17 marks)

Q31.

(a) Electra is employed by E & G Ltd to install electricity meters in new houses on an estate. Her time, X minutes, to install a meter may be assumed to be normally distributed with a mean of 48 and a standard deviation of 20.

Determine:

(i) P(X < 60);

(2)

(ii) P(30 < X < 60);

(3)

(iii) the time, k minutes, such that P(X < k) = 0.9.

(4)

(b) Gazali is employed by E & G Ltd to install gas meters in the same new houses. His time, Y minutes, to install a meter has a mean of 37 and a standard deviation of 25.

(i) Explain why Y is unlikely to be normally distributed.

(2)

(ii) State why $\overline{Y}$, the mean of a random sample of 35 gas meter installations, is likely to be approximately normally distributed.

(1)

(iii) Determine P($\overline{Y}$ > 40).

(4)

(Total 16 marks)

Q32.

(a) The baggage loading time, X minutes, of a chartered aircraft at its UK airport may be modelled by a normal random variable with mean 55 and standard deviation 8.

Determine:

(i) P(X < 60);

(3)

(ii) P(55 < X < 60).

(2)

(b) The baggage loading time, Y minutes, of a chartered aircraft at its overseas airport may be modelled by a normal random variable with mean μ and standard deviation 16.

Given that P(Y < 90) = 0.95, find the value of μ.

(4)

(Total 9 marks)

Q33.

(a) The weight, X grams, of soup in a carton may be modelled by a normal random variable with mean 406 and standard deviation 4.2.

Find the probability that the weight of soup in a carton:

(i) is less than 400 grams;

(3)

(ii) is between 402.5 grams and 407.5 grams.

(4)

(b) The weight, Y grams, of chopped tomatoes in a tin is a normal random variable with mean μ and standard deviation σ.

(i) Given that P(Y < 310) = 0.975, explain why:

$$310 - \mu = 1.96\sigma$$

(3)

(ii) Given that P(Y < 307.5) = 0.86, find, to two decimal places, values for μ and σ.

(4)

(Total 14 marks)

Q34.

The heights of sunflowers may be assumed to be normally distributed with a mean of 185 cm and a standard deviation of 10 cm.

Determine the probability that the height of a randomly selected sunflower:

(a) is less than 200 cm;

(3)

(b) is more than 175 cm;

(3)

(c) is between 175 cm and 200 cm.

(2)
(Total 8 marks)

Q35.

Chopped lettuce is sold in bags nominally containing 100 grams.

The weight, X grams, of chopped lettuce, delivered by the machine filling the bags, may be assumed to be normally distributed with mean μ and standard deviation 4.

(a) Assuming that μ = 106, determine the probability that a randomly selected bag of chopped lettuce:

(i) weighs less than 110 grams;

(3)

(ii) is underweight.

(3)

(b) Determine the minimum value of μ so that at most 2 per cent of bags of chopped lettuce are underweight. Give your answer to one decimal place.

(4)
(Total 10 marks)